

# Technical Document #478: Software Engineering - Comprehensive Overview

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Domain-driven design (DDD) models software around business domains. It improves communication between technical and business stakeholders.

Test-driven development (TDD) writes tests before writing code. It improves code quality and design through small, incremental changes.

Microservices architecture structures applications as independently deployable services. Each service owns its data and business logic.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

### Key Takeaways:

- Code review examines code changes before merging.
- Test-driven development (TDD) writes tests before writing code.
- Continuous deployment (CD) automatically deploys code changes to production.